KAIZEN MATHS

# Lesson starter

**Remainder and Factor Theorem**

Remainder Theorem

Name _____

**S1**

Find the remainder when $P(x) = 3x^3 - x + 1$ is divided by $x - 1$.

**S2**

Find the remainder when $P(x) = 2x^3 - 3x^2 - 5x - 2$ is divided by $x + 5$.

**S3**

Find the remainder when $P(x) = 4x^2 + 5x - 5$ is divided by $2x - 3$.

**S4**

For $P(x) = x^2 + px + q$, remainders for $x + 3$ and $x - 3$ are $17$ and $-7$. Find $p, q$.



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# Exit ticket

**Remainder and Factor Theorem**

Remainder Theorem

Name _____

**E1**

Find the remainder when $P(x) = 4x^3 - 2x^2 + 3x + 4$ is divided by $x + 2$.

**E2**

Find the remainder when $P(x) = 2x^2 + 5x$ is divided by $x + 4$.

**E3**

Find the remainder when $P(x) = 2x^2 + 5x + 2$ is divided by $3x - 2$.



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Find the remainder when $P(x) = 2x^2 + 5x + 2$ is divided by $3x - 2$.

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# Teacher answers & working

**Remainder and Factor Theorem · Remainder Theorem**

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## S1

Find the remainder when $P(x) = 3x^3 - x + 1$ is divided by $x - 1$.

**Answer:**

$R = P(1) = 3$

**Working:**

Step 1: Use the remainder theorem: when dividing by $x - a$, the remainder is $P(a)$.

Step 2: The divisor is x - 1, so the value to substitute is $x = 1$.

Step 3: Substitute $1$ into the polynomial.

$P(1) = 3(1)^3 - (1) + 1$

$P(1) = 3$

Step 4: The evaluated value is the remainder.

$\boxed{R = 3}$

## S2

Find the remainder when $P(x) = 2x^3 - 3x^2 - 5x - 2$ is divided by $x + 5$.

**Answer:**

$R = P(-5) = -302$

**Working:**

Step 1: Set the divisor equal to zero: $x + 5 = 0$.

Step 2: This gives $x = -5$. That is the value used in the remainder theorem.

Step 3: Evaluate $P(-5)$.

$P(-5) = 2(-5)^3 - 3(-5)^2 - 5(-5) - 2$

$P(-5) = -302$

Step 4: The remainder is $-302$.

$\boxed{R = -302}$

# Teacher answers & working

**Remainder and Factor Theorem · Remainder Theorem**

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## S3

Find the remainder when $P(x) = 4x^2 + 5x - 5$ is divided by $2x - 3$.

**Answer:**

$R = P\left(\frac{3}{2}\right) = \frac{23}{2}$

**Working:**

Step 1: Set the divisor equal to zero: $2x - 3 = 0$.

Step 2: Solve for $x$: $x = \frac{3}{2}$.

Step 3: Use the remainder theorem, so $R = P\left(\frac{3}{2}\right)$.

$P(\frac{3}{2}) = 4(\frac{3}{2})^2 + 5(\frac{3}{2}) - 5$

$P(\frac{3}{2}) = \frac{23}{2}$

Step 4: The remainder is $\frac{23}{2}$.

## S4

For $P(x) = x^2 + px + q$, remainders for $x + 3$ and $x - 3$ are $17$ and $-7$. Find $p, q$.

**Answer:**

$p = -4,\ q = -4$

**Working:**

Use $P(-3) = 17$ and $P(3) = -7$.

$-3p + q = 8$

$3p + q = -16$

Solving gives $p = -4$ and $q = -4$.

## E1

Find the remainder when $P(x) = 4x^3 - 2x^2 + 3x + 4$ is divided by $x + 2$.

**Answer:**

$R = P(-2) = -42$

**Working:**

Step 1: Use the remainder theorem: when dividing by $x - a$, the remainder is $P(a)$.

Step 2: The divisor is x + 2, so the value to substitute is $x = -2$.

Step 3: Substitute $-2$ into the polynomial.

$P(-2) = 4(-2)^3 - 2(-2)^2 + 3(-2) + 4$

$P(-2) = -42$

Step 4: The evaluated value is the remainder.

$\boxed{R = -42}$

# Teacher answers & working

**Remainder and Factor Theorem · Remainder Theorem**

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## E2

Find the remainder when $P(x) = 2x^2 + 5x$ is divided by $x + 4$.

**Answer:**

$R = P(-4) = 12$

**Working:**

Step 1: Set the divisor equal to zero: $x + 4 = 0$.

Step 2: This gives $x = -4$. That is the value used in the remainder theorem.

Step 3: Evaluate $P(-4)$.

$P(-4) = 2(-4)^2 + 5(-4)$

$P(-4) = 12$

Step 4: The remainder is $12$.

$\boxed{R = 12}$

## E3

Find the remainder when $P(x) = 2x^2 + 5x + 2$ is divided by $3x - 2$.

**Answer:**

$R = P\left(\frac{2}{3}\right) = \frac{56}{9}$

**Working:**

Step 1: Set the divisor equal to zero: $3x - 2 = 0$.

Step 2: Solve for $x$: $x = \frac{2}{3}$.

Step 3: Use the remainder theorem, so $R = P\left(\frac{2}{3}\right)$.

$P(\frac{2}{3}) = 2(\frac{2}{3})^2 + 5(\frac{2}{3}) + 2$

$P(\frac{2}{3}) = \frac{56}{9}$

Step 4: The remainder is $\frac{56}{9}$.